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by

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We hereby declare that the project report entitled **"Improved Fungal Growth Optimization (IFGO) and its use in Clustering Improvement"** is an authentic record of our own work carried out at the **Department of Computer Science and Engineering, Techno India University, West Bengal**, during the **8th semester of the academic year 2025-2026** under the supervision of **Prof. Kalyan Kumar Das**, Assistant Professor.

We further declare that the matter embodied in this project report has not been submitted by us for the award of any other degree or diploma of this or any other Institute/University. All the information have been obtained and presented in accordance with academic rules and ethical conduct. We also declare that, as required by these rules and conduct, we have fully cited and referenced all materials and results that are not original to this work.

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ABSTRACT

This study presents a Improved Fungal Growth Optimization (IFGO) technique, an improved version of the nature-inspired Fungal Growth Optimization (FGO) algorithm. The proposed method copies fungal behaviors such as hyphal growth, chemotropism, branching and spore germination. IFGO incorporates chaotic initialization, Levy flight-based mutation, opposition-based learning and an adaptive exploration–exploitation mechanism. A composite multi-objective fitness function is implemented using multiple cluster validity indices to extract improved clustering performance. The proposed algorithm is also hybridized with traditional clustering methods like K-Means and Fuzzy C-Means (FCM) for enhancement of centroid optimization. In addition, IFGO is evaluated against several metaheuristic algorithms using five clustering validity indices and multiple statistical analyses. IFGO outperforms or is competitive with all of the algorithms chosen for the experimental study. This implies that IFGO is a high-performing optimizer and a powerful alternative technique for clustering related complex optimization problems.

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Chapter 1

Introduction

1.1 Motivation

Clustering [1] is a fundamental unsupervised learning method for identifying inherent structures present within unlabeled data. It is a vital component in various applications, including pattern recognition [2], image segmentation [3] and data analysis where it simplifies complex data to facilitate efficient decision-making. However, the traditional clustering methods [4] including K-Means and Fuzzy C-Means (FCM), possess certain disadvantages. They are extremely sensitive to the initialization of cluster centroids, which often leads to premature convergence to local minima. In addition, the assumption of spherical cluster structures is considered one of the major drawbacks of these classical methods. The requirement of a fixed number of clusters limits their applicability. The pre-defined number of clusters might result in an incorrect representation of the data.

1.2 Objectives

To overcome these shortcomings, the present research study aims at introducing a new form of FGO method called Improved Fungal Growth Optimization (IFGO).

The main contributions of this research can be summarized as follows:

- An improved balance between exploration and exploitation phases.

- The integration of chaotic initialization, Levy flight-based mutation and opposition-based learning to enhance global search capability and convergence stability.
- The development of a composite multi-objective fitness function that combines multiple cluster validity indices for robust and balanced clustering.
- A novel hybrid clustering framework that integrates IFGO with K-Means and FCM for effective centroid refinement.
- Extensive experimental and statistical validation on benchmark datasets demonstrating the superiority of the proposed method.

1.3 Organization

In order to solve these issues, researchers have developed new hybrid methods by integrating the use of meta-heuristic [5] optimization methods and conventional clustering techniques. Amongst all such hybrid methods, Fungal Growth Optimization (FGO) [6] technique exhibits promising results due to the inherent adaptive search nature of the algorithm. Nevertheless, there are various disadvantages with the traditional FGO method. The main contributions of this research can be summarized as follows:

- An improved balance between exploration and exploitation phases.
- The integration of chaotic initialization, Levy flight-based mutation and opposition-based learning to enhance global search capability and convergence stability.
- The development of a composite multi-objective fitness function that combines multiple cluster validity indices for robust and balanced clustering.
- A novel hybrid clustering framework that integrates IFGO with K-Means and FCM for effective centroid refinement.
- Extensive experimental and statistical validation on benchmark datasets demonstrating the superiority of the proposed method.

Chapter 2

Project Timeline

TABLE 2.1: Gantt Chart Breakdown of Project Development

Milestone	Task Description	Jan	Feb	Mar	Apr	May
M1	Selection Algorithms	■				
M2	Algorithm Evaluation		■■			
M3	Finalization		■			
M4	Algorithm Development			■■		
M5	Dataset Selection			■		
M6	Result (Graphs & Plots)				■■	
M7	Paper Writing				■	■

Chapter 3

Related Work

3.1 Centroid Initialization Strategies

Cluster analysis has been extensively used for different applications in machine learning [7] and data mining [8]. Among all clustering techniques, K-Means [9] and Fuzzy C-Means [10] clustering are the most popular and computationally efficient techniques. These conventional methods exhibit notable shortcomings. For instance, the initial values of parameters and local optimum convergence are major limitations of the mentioned clustering techniques.

Different metaheuristic techniques have been integrated with these traditional clustering techniques to overcome the limitations. The metaheuristic techniques selected for this study include : Genetic Algorithm Optimization (GA) [11], Particle Swarm Optimization (PSO) [12], Gorilla Troop Optimization (GTO) [13], Billiards Optimization (BO) [14], Improved Social Group Optimization (ISGO) [15], Energy Optimization (EO) [16], Hippo Optimization (HO) [17], Meta-OOD [18], Sequential Pattern Mining Optimization (SPM) [19] and General Trend (GT) [20]. The existing metaheuristic-based hybrid clustering algorithms are facing major challenges like low convergence rate, high computational complexity and exploration-exploitation balance despite of the improvements.

3.2 Selected Metaheuristic Optimization

3.2.1 Genetic Algorithm Optimization (GA)

Genetic Algorithm Optimization (GA) is a robust, population-based global search metaheuristic inspired by the principles of natural selection and evolutionary genetics. In cluster analysis, GA is widely utilized to optimize center-based clustering by simultaneously processing a diverse population of candidate solutions, represented as chromosomes encoding potential cluster centroids. Through iterative applications of selection, crossover, and mutation operators, the algorithm effectively navigates high-dimensional search spaces to minimize intra-cluster variance without requiring gradient information. However, standard GA exhibits structural limitations which includes slow late-stage convergence and high computational complexity. Researchers hybridize it with techniques like K-Means to stabilize the exploration-exploitation trade off and avoid local optima stagnation.

TABLE 3.1: Comparative Analysis of Genetic Algorithm (GA) Variants in Clustering Optimization

GA Variant	**Core Modification Mechanic**	**Targeted Clustering Challenge**
Real-Coded GA (RCGA)	Utilizes continuous floating-point vectors instead of binary strings for chromosome encoding.	Eliminates quantization errors and preserves exact spatial coordinates of centroids.
Adaptive GA (AGA)	Dynamically scales crossover and mutation probabilities based on population fitness variance.	Prevents premature convergence in early generations and promotes fine local exploitation later.
Hybrid GA-KMeans	Seeds classical K-Means with the finest global chromosomes produced during the GA search.	Mitigates slow late-stage convergence by combining global exploration with fast local refinement.
Niching / Multi-Modal GA	Implements sharing or crowding distances to maintain structural diversity among sub-populations.	Prevents the algorithm from converging onto a single dominant local optimum in multimodal spaces.

3.2.2 Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO) is a population based metaheuristic inspired by the social behaviors of bird flocks and fish schools. In centroid based clustering, the system initializes a swarm of candidate solutions termed as particles where position of each particle vector represents a complete set of cluster centers. Particles adjust their velocities dynamically by moving through a high dimensional search space, based on their individual historical best position (*pbest*) and the overall global best position (*gbest*). This collective intelligence provides rapid convergence and efficient global exploration without asking for gradient information. However, standard PSO remains highly susceptible to premature convergence and stagnation in local optima when dealing with complex data landscapes as the loss of swarm diversity often freezes particles before achieving local refinement.

TABLE 3.2: Core Structural Components of PSO in Clustering Applications

Component	**Algorithmic Function**	**Impact on Clustering**
Velocity Vector	Directs the step size and trajectory changes of centroids across iterations.	Governs the rate of exploration and exploitation.
Cognitive Memory	Pulls the particle back toward its personal historically best-performing clustering configuration.	Preserves individual localized knowledge.
Social Component	Attracts the entire swarm toward the best global centroid matrix found so far.	Accelerates convergence across the search space.
Inertia Weight	Moderates the influence of previous velocity vectors on current movements.	Balances wide-area scanning with fine tuning.

3.2.3 Gorilla Troop Optimization (GTO)

Gorilla Troop Optimization (GTO) is a nature inspired metaheuristic algorithm that copies the social intelligence hierarchical troop dynamics of gorillas [13]. In data clustering, candidate centroid configurations are represented as individual gorilla where search operations are strictly governed by the positioning of silver back. The algorithm operates through an exploration phase simulating migration to new environments and moving toward unknown coordinates. An exploitation phase which models close range silver back monitoring and competition for resources. This structure grants it a distinct advantage in tracking dynamic search terrains and escaping early local traps. However when applied to high dimensional clustering it suffers from a high computational overhead.

TABLE 3.3: Algorithmic Phases of GTO and Their Roles in Centroid Refinement

Phase	**Behavioral Mechanism**	**Impact on Clustering**
Exploration (Migration)	Moving toward random locations or tracking known group members.	Discovers entirely new coordinate zones for centroids.
Exploration (Transfer)	Migrating the troop toward a completely fresh environment.	Enhances population diversity to bypass local traps.
Exploitation (Follower)	Following the dominant silverback gorilla's spatial vector.	Accelerates local convergence toward optimal clusters.
Exploitation (Competition)	Group members competing for resources and space.	Performs fine-grained spatial tuning of cluster centers.

3.2.4 Billiards Optimization (BO)

Billiards Optimization (BO) Algorithm is a physics and game based metaheuristic that mathematically models the kinematics, vector mechanics and strategic rules of a classic billiards game [14]. Within center based data clustering, candidate centroid configurations are treated as multi dimensional billiard balls distributed across the search landscape while the highest performing configurations are designated as pockets. The optimization engine updates positions using momentum preservation formulas and vector algebra to simulate the elastic collisions that occur when balls strike one another or the table borders. This collision mechanic helps the algorithm escape local optimum convergence by scattering stagnated particles and exploring wide regions of the coordinate space. However, when clustering high-dimensional datasets, it can suffer from high computational complexity per iteration due to intensive vector velocity calculations along with a drop in final convergence speed as balls struggle to settle precisely into optimal pocket coordinates.

TABLE 3.4: Kinematic Mechanics of BOA and Their Roles in Centroid Placement

Mechanical Element	**Mathematical Definition**	**Impact on Clustering**
Billiard Balls	Dynamic vectors holding candidate coordinates.	Scans the search space to find ideal cluster centroids.
Table Pockets	Stationary reference states based on the top eight local solutions.	Pulls the population toward high-density regions.
Ball Collisions	Kinetic energy transfer and trajectory displacement.	Maintains population diversity to bypass local traps.
Vector Algebra Rules	Velocity updates dictated by angle of reflection and cue force.	Regulates the balance between wide exploration and fine exploitation.

3.2.5 Improved Social Group Optimization (ISGO)

Improved Social Group Optimization (ISGO) is a human centric metaheuristic algorithm that models the behavioral patterns of individuals acquiring and sharing knowledge within a social structure [15]. In clustering applications each candidate centroid configuration is treated as an individual person within a group. Their knowledge level corresponds to their structural clustering fitness. The algorithm progresses through an updated learning phase where group members improve their knowledge by interacting with a globally superior individual and an improved social phase where individuals share localized insights with random peers. This twin layer communication structure allows to avoid early population homogeneity and bypass local optimum trapping. However, when tackling high-dimensional datasets the algorithm can suffer from slow convergence in late stage iterations due to its high reliance on interaction variables rather than fine grained local gradient exploitation.

TABLE 3.5: Knowledge-Sharing Phases of ISGO and Their Roles in Centroid Placement

Phase	**Social Mechanism**	**Impact on Clustering**
Improved Learning Phase	Direct knowledge acquisition from the best-performing teacher vector.	Drives rapid global convergence toward high-quality cluster zones.
Improved Social Phase	Multi-directional knowledge sharing among randomly selected peers.	Maintains spatial diversity across candidates to prevent early trapping.
Self-Improvement Operator	Individual refinement of ideas using localized introspection.	Tunes coordinate dimensions to enhance cluster compactness.
Dynamic Tuning Coefficients	Adaptive scaling parameters governing individual step sizes.	Coordinates the transition from global scanning to fine exploitation.

3.2.6 Energy Optimization (EO)

Energy Optimization (EO) technique particularly physics-based models like the Equilibrium Optimizer or thermodynamic inspired frameworks, represent a highly efficient class of metaheuristics utilized to resolve centroid-based clustering traps [16]. In cluster analysis candidate solutions are modeled as physical states or particles moving through a search space governed by minimization forces. The objective of the system is to reduce global network "energy". The algorithm models dynamic state transitions, allowing candidate centroids to adaptively fluctuate between high energy exploration phases and low energy local exploitation sinks. This thermodynamic balancing act helps the optimization engine systematically escape premature stagnation and local optima locking. However, when confronting massive and high dimensional datasets standard EO algorithms can struggle with high computational complexity per iteration due to continuous concentration or energy state updates ; which occasionally results in a slow final convergence rate during the exact positioning of cluster boundaries.

TABLE 3.6: Thermodynamic Mechanics of EO and Their Roles in Centroid Placement

Thermodynamic Element	**Algorithmic Formulation**	**Impact on Clustering**
Energy States	Vectors representing the coordinate placement of candidate cluster centroids.	Guides the shifting trajectories of cluster centers.
Equilibrium Candidates	The top globally performing solutions used as focal reference states.	Directs the population toward high-density regions.
High-Energy Phase	Expanded step sizes simulating chaotic particle interactions.	Drives wide global exploration to bypass local traps.
Low-Energy Phase	Decaying step sizes mimicking thermal cooling and stabilization.	Executes fine-grained spatial tuning of cluster coordinates.

3.2.7 Hippo Optimization (HO)

Hippopotamus Optimization (HO) is a recently introduced swarm intelligence metaheuristic modeled after the spatial ecology, social hierarchy and defensive survival tactics of wild hippopotamuses [17]. When applied to center based data clustering, candidate centroid matrices are mapped as individual hippos within a herd whose movements are regulated by a dominant male leader. The optimization framework balances exploration and exploitation through three primary behavioral stages. Simulating herd position updates in aquatic territories, executing aggressive defense mechanisms against local predators and utilizing active avoidance maneuvers. While these distinct behavioral loops enable the algorithm to handle non-linear search landscapes smoothly and escape structural local trapping, standard HO still faces limitations. Encountering high dimensional clustering spaces, its performance can degrade due to single range parameter boundaries, creating a tendency toward slow late-stage convergence and high computational complexity.

TABLE 3.7: Ecological Stages of HO and Their Functional Roles in Clustering

Ecological Stage	**Behavioral Formulation**	**Impact on Clustering**
Herd Position Shifting	Simulates hippos updating their positions relative to the dominant male and herd averages.	Drives broad exploration to locate initial cluster zones.
Territorial Defense	Models aggressive responses and charge trajectories toward perceived environmental threats.	Accelerates search velocity toward high-performing centroids.
Predator Avoidance	Implements stochastic displacement vectors to actively evade hunting predators.	Generates coordinate diversity to break out of local optima traps.
Dominance Updating	Continuous reassessment of individual fitness to dynamically update the lead male hippo.	Pulls the remaining population toward the current best global layout.

3.2.8 Meta-OOD

Meta-OOD represents an advanced framework within the paradigm of out of distribution (OOD) detection, combining the concepts of meta learning with domain generalization to isolate anomalous or distribution-shifted samples [18]. In the context of center-based cluster analysis ; Meta-OOD is utilized to establish an adaptive model of normality enabling the system to distinguish between core in distribution data points and virtual outlier vectors. Instead of assuming static margins, Meta-OOD treats variations in cluster densities as sub tasks, training a generalized boundary evaluator that quickly adapts to shifting spatial layouts. While this technique provides superior protection against data noise and outlier corruption, integrating it into optimization pipelines presents unique structural difficulties. When handling high dimensional data, Meta-OOD introduces substantial computational complexity per iteration, along with high sensitivity to boundary threshold settings which can stall the overall optimization engine and degrade the final convergence speed of centroid refinement.

TABLE 3.8: Structural Mechanics of Meta-OOD in Boundary Selection

Component	**Operational Formulation**	**Impact on Clustering**
Meta-Representation	Learns generalized structural features across diverse sub-distributions.	Provides stable, domain-invariant centroid boundaries.
Virtual OOD Generator	Synthesizes pseudo-outliers based on edge coordinates of the input data.	Calibrates decision margins to eliminate background noise.
Lightweight Adapter	Performs fast updates to classification heads using few reference samples.	Enhances system adaptability to dynamically shifting datasets.
Energy Regularization	Measures free-energy distribution scores to detect data anomalies.	Prevents local optima traps caused by outlier data points.

3.2.9 Sequential Pattern Mining Optimization (SPM)

Sequential Pattern Mining Optimization (SPM) combines combinatorial pattern matching with heuristic search engines to discover valid subsequences in transaction databases or ordered series [19]. Adapted to centroid-based data clustering the optimization engine shifts from identifying spatial clusters to evaluating sequential associations and mapping candidate sequences into discrete solution matrices. The algorithm evaluates candidate configurations using support, confidence and leverage constraints, optimizing search bounds to trim unpromising branches early. This systematic pruning makes the algorithm highly efficient at handling structured sequences and navigating large search paths. However, when applied to high-dimensional datasets SPM algorithms face combinatorial explosion due to a massive increase in candidate sequences creating high computational complexity per iteration, memory overhead, and a slow final convergence rate when trying to refine exact centroid strings.

TABLE 3.9: Structural Phases of SPM Optimization and Their Roles in Clustering

Operational Element	**Mathematical Formulation**	**Impact on Clustering**
Sequence Encoding	Represents coordinate trajectories as ordered, discrete symbolic strings.	Standardizes chronological data for systematic evaluation.
Support Filtering	Evaluates candidate recurrence frequencies against a predefined minimum boundary.	Trims sparse coordinate patterns to speed up global scanning.
Prefix Tree Pruning	Uses anti-monotonic properties to eliminate unviable downstream subbranches.	Reduces execution times by shrinking the active search area.
Constraint Updates	Adaptively alters validation thresholds based on population density changes.	Directs the population toward high-density regions to fine-tune boundaries.

3.2.10 General Trend (GT)

General Trend (GT) optimization is a modern data driven metaheuristic that models the directional trajectories and momentum shifts of macro-scale operational data to locate global optima [20]. In center based clustering applications, GT abstracts candidate centroid configurations as moving states governed by prevailing mathematical trends observed across the historical performance of the population. The optimization engine updates its positions by analyzing historical success rates, applying trend following operators to accelerate search velocity along fruitful paths while using inversion vectors to deviate from downward trends. This predictive framework provides strong guidance through complex, multi modal search terrains, ensuring steady progress toward high density regions. However, when applied to high dimensional problems, it can exhibit sensitivity to initial historical window sizes which occasionally induces a slow late stage convergence rate and high computational complexity when fine tuning exact cluster center coordinates.

TABLE 3.10: Operational Mechanisms of the GT Optimizer in Cluster Analysis

Mechanism	**Mathematical Formulation**	**Impact on Clustering**
Trend Extraction	Evaluates the mathematical slope of population fitness changes over successive generations.	Establishes a highly directed trajectory for moving centroids.
Trend-Following Factor	Generates proportional step updates aligned with historical improvement vectors.	Accelerates local convergence toward high-quality cluster zones.
Trend Inversion Operator	Executes a directional shift when performance hits stagnation thresholds.	Breaks up population stagnation to bypass local optima traps.
Historical Memory Window	Maintains a sliding-scale matrix recording the best coordinates from previous runs.	Balances wide exploratory scanning with precise centroid adjustments.

3.3 Advancements of Fungi-Inspired Metaheuristics

A relatively new algorithm, Fungal Growth Optimization (FGO) has shown promising results. Fungal Growth Optimization is a nature-inspired metaheuristic algorithm that emulates fundamental fungal growth mechanisms including hyphal-growth, chemotropism, branching, and spore germination. Hyphal-growth replicated hyphal extension to precisely explore the search space. Chemotropism helps to reach and exploit nutrient rich regions inside the search space, promoting performance against stagnation into local optima and slow convergence speed. Branching behavior replicates the way how new hyphal branches form next to an existing hypha explore regions for searching of more nutrient. Branching behavior promotes the exploratory operator. Germination represents how existing hyphae explore new environment to reach safer and nutrient rich areas. Due to its adaptive exploration capability and progressive refinement of candidate solutions, Fungal Growth Optimization (FGO) has demonstrated improved reliability compared to many conventional metaheuristic approaches.

To further enhance its performance, several variants of FGO have been introduced in recent studies. For instance, Bioluminescent Fungi Optimization (BFO) [31] incorporates a light-based attraction mechanism that enhances population diversity and improves the exploration capability of the algorithm. Similarly, Fungi Kingdom Expansion (FKE) [32] employs deterministic expansion strategies, combining immobile expansion for local search and mobile expansion for global search, thereby achieving a more balanced exploration–exploitation trade-off.

FGO still suffers from limitations. Such as high computational complexity, sensitivity to parameter settings and slow convergence in high-dimensional problems. It restricts it's effectiveness for large scale real life applications. Furthermore, adaptive parameter tuning [21], opposition based learning [22], Lévy flight strategies [23] and hybridization [24] with traditional algorithms have been explored to improve the overall performance of Fungal Growth Optimization. These enhancements are introduced in IFGO for strengthening the balance and to propose a better alternative algorithm.

3.4 Gap Identification

Although clustering has been extensively studied in machine learning and data mining several limitations still exist in traditional and metaheuristic based clustering approaches. To address these issues there is a need for an improved clustering framework that enhances centroid initialization, strengthens global search capability and provides better robustness across diverse datasets. The proposed Improved Fungal Growth Optimization (IFGO) based clustering approach aims to address these gaps by integrating chaotic initialization, adaptive search strategies, Lévy flight mutation and hybrid clustering mechanisms to achieve more stable clustering performance. To overcome these problems, the present study aims at introducing a new form of FGO method called Improved Fungal Growth Optimization (IFGO).

TABLE 3.11: Summary of Main Research Contributions

Core Focus	**Specific Mechanism**	**Primary Benefit**
Search Dynamics	Balancing exploration and exploitation phases	Improved optimization behavior
Algorithmic Enhancements	Integration of chaotic initialization, Lévy flight-based mutation, and opposition-based learning	Enhanced global search capability and convergence stability
Fitness Function	Development of a composite multi-objective fitness function combining multiple cluster validity indices	Robust and balanced clustering performance
Clustering Framework	Hybridizing IFGO with K-Means and FCM	Effective centroid refinement
Validation	Extensive experimental and statistical testing on benchmark datasets	Demonstrated superiority of the proposed method

Chapter 4

Methodology

4.1 Algorithm

Algorithm 1 Improved-FGO Based Clustering (IFGO)

Input: Dataset X, clusters K, population P, iterations T

Output: Optimal centroids Z

Initialize bounds lb, ub from X

Generate initial population of candidate solutions

Evaluate initial fitness $F(Z)$

Assign samples to nearest centroid

Compute clustering fitness using validity indices

Update best solution Z

Apply adaptive parameter control

Apply population-based search strategy

Apply exploration and exploitation mechanisms

Ensure boundary constraints

Perform local refinement using clustering methods:

return Z

Algorithm 1 illustrates the workflow of the proposed IFGO algorithm.

4.2 Mathematical Formulation

4.2.1 Problem Definition

Let the dataset be represented as:

$$X = \{x_1, x_2, \ldots, x_N\}, \quad x_i \in \mathbb{R}^d \tag{4.1}$$

where N denotes the total number of data samples and d represents the feature dimensionality.

The clustering objective is to partition X into K disjoint clusters by determining optimal centroids:

$$C = \{c_1, c_2, \ldots, c_K\}, \quad c_j \in \mathbb{R}^d \tag{4.2}$$

In the proposed Improved-FGO framework, each candidate solution encodes all cluster centroids as:

$$Z_i = [c_1, c_2, \ldots, c_K] \in \mathbb{R}^{K \times d} \tag{4.3}$$

The clustering task is therefore formulated as a constrained optimization problem:

$$\max_{Z} \; F(Z) \tag{4.4}$$

subject to:

$$c_j \in [lb, ub]^d \tag{4.5}$$

where lb and ub denote lower and upper bounds derived from the dataset feature space.

4.2.2 Cluster Assignment Mechanism

For a given solution Z, each data sample is assigned to its nearest centroid using Euclidean distance formula:

$$\ell_i = \arg \min_{j \in \{1,\ldots,K\}} \|x_i - c_j\|_2 \tag{4.6}$$

This assignment produces K clusters $C_1, C_2, \ldots, C_K$.

4.2.3 Composite Fitness Function

To enhance clustering robustness and structural validation, a multi-objective fitness function is constructed by integrating five validity indices:

$$F = S + \frac{1}{1+DB} + \frac{CH}{1+CH} + D + \frac{1}{1+XB} \tag{4.7}$$

Here, DB and XB are transformed into maximization form.

For intermediate iterations, a computationally efficient proxy objective is employed:

$$F = -\sum_{i=1}^{N} \min_{j} \|x_i - c_j\| \tag{4.8}$$

4.2.4 Silhouette Index

$$S = \frac{1}{N} \sum_{i=1}^{N} \frac{b(i) - a(i)}{\max(a(i),\, b(i))} \tag{4.9}$$

where $a(i)$ denotes the intra-cluster distance and $b(i)$ represents the nearest-cluster distance.

4.2.5 Davies–Bouldin Index

$$DB = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{\|c_i - c_j\|} \right) \tag{4.10}$$

where σ_i is the average intra-cluster dispersion of cluster i.

4.2.6 Calinski–Harabasz Index

$$CH = \frac{\mathrm{Tr}(B_K)/(K-1)}{\mathrm{Tr}(W_K)/(N-K)} \tag{4.11}$$

where B_K and W_K represent the between-cluster and within-cluster scatter matrices, respectively.

4.2.7 Dunn Index

$$D = \frac{\min_{i \neq j} \delta(C_i, C_j)}{\max_k \Delta(C_k)} \tag{4.12}$$

where $\delta(C_i, C_j)$ denotes the inter-cluster distance and $\Delta(C_k)$ represents the cluster diameter.

4.2.8 Xie–Beni Index

$$XB = \frac{\sum_{j=1}^{K} \sum_{x_i \in C_j} \|x_i - c_j\|^2}{N \cdot \min_{i \neq j} \|c_i - c_j\|^2} \tag{4.13}$$

4.2.9 Chaotic Initialization

Population diversity is enhanced using logistic chaotic mapping:

$$r_{t+1} = 4r_t(1 - r_t) \tag{4.14}$$

Initial positions are generated as:

$$Z_{i,j} = lb + r_t(ub - lb) \tag{4.15}$$

4.2.10 Adaptive Position Update Strategy

To balance exploration and exploitation, an adaptive coefficient is defined as:

$$r(t) = r_{\max} - \left(\frac{t}{T}\right)^{\beta} (r_{\max} - r_{\min}) \tag{4.16}$$

where T denotes the maximum number of iterations.

The fungi position update rule is expressed as:

$$Z_i^{(t+1)} = Z_i^t + r(t)\left(Z_{\text{best}}^t - Z_i^t\right) \tag{4.17}$$

4.2.11 Lévy Flight Exploration

To prevent premature convergence, Lévy perturbation is introduced:

$$Z_i^{(t+1)} = Z_i^{(t+1)} + \alpha \cdot \text{Levy}(\lambda) \tag{4.18}$$

4.2.12 Opposition-Based Learning

For poorly performing solutions, opposition learning is applied:

$$Z_i^{\text{new}} = lb + ub - Z_i \tag{4.19}$$

4.2.13 K-Means Update

$$c_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i \tag{4.20}$$

4.2.14 Fuzzy C-Means Update

Membership:

$$u_{ij} = \frac{\left(\frac{1}{\|x_i - c_j\|}\right)^{\frac{2}{m-1}}}{\sum_{k=1}^{K} \left(\frac{1}{\|x_i - c_k\|}\right)^{\frac{2}{m-1}}} \tag{4.21}$$

Centroid update:

$$c_j = \frac{\sum_{i=1}^{N} u_{ij}^m x_i}{\sum_{i=1}^{N} u_{ij}^m} \tag{4.22}$$

4.2.15 Final Optimization Model

The overall IFGO based clustering model is defined as:

$$\max_{Z \in [lb,\, ub]^{K \times d}} F(Z) \tag{4.23}$$

Chapter 5

Mathematical Expression of Related Algorithms

5.1 Fungal Growth Optimization (FGO)

Fungal Growth Optimization (FGO) is a bio inspired optimization algorithm that mimics how fungi grow, spread and search for nutrients in nature. This is chosen as our standard algorithm. For pure FGO algorithm the mathematical formation is given below.

5.1.1 Population Initialization

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{id}) \tag{5.1}$$

$$x_{ij} = x_j^{\min} + rand(0,1) \times (x_j^{\max} - x_j^{\min}) \tag{5.2}$$

5.1.2 Fitness Evaluation

$$f_i = f(X_i) \tag{5.3}$$

5.1.3 Spore Dispersal (Exploration Phase)

$$X_i^{t+1} = X_i^t + r_1 \times (X_{best}^t - X_i^t) \tag{5.4}$$

5.1.4 Mycelium Expansion (Local Exploitation)

$$X_i^{t+1} = X_i^t + r_2 \times (X_j^t - X_k^t) \tag{5.5}$$

5.1.5 Nutrient Absorption

$$X_i^{t+1} = X_i^t + r_3 \times (X_{best}^t - X_{worst}^t) \tag{5.6}$$

5.1.6 Adaptive Growth Factor

$$G(t) = G_0 \times \left(1 - \frac{t}{T}\right) \tag{5.7}$$

5.1.7 Position Update with Growth

$$X_i^{t+1} = X_i^t + G(t) \times rand(0,1) \times (X_{best}^t - X_i^t) \tag{5.8}$$

5.1.8 Boundary Control

$$x_{ij} = \begin{cases} x_j^{\min}, & x_{ij} < x_j^{\min} \\ x_j^{\max}, & x_{ij} > x_j^{\max} \end{cases} \tag{5.9}$$

5.2 Genetic Algorithm (GA)

Genetic Algorithm (GA) is a search technique inspired by natural selection that mimics evolution—using selection, crossover, and mutation—to find optimal solutions to complex problems.

5.2.1 Population Initialization

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{id}) \tag{5.10}$$

$$x_{ij} = x_j^{\min} + rand(0,1) \times (x_j^{\max} - x_j^{\min}) \tag{5.11}$$

5.2.2 Fitness Evaluation

$$f_i = f(X_i) \tag{5.12}$$

5.2.3 Selection (Roulette Wheel Mechanism)

$$P_i = \frac{f_i}{\sum_{k=1}^{N} f_k} \tag{5.13}$$

5.2.4 Crossover (Exploration Phase)

$$X_i^{t+1} = r_1 \times X_a^t + (1 - r_1) \times X_b^t \tag{5.14}$$

5.2.5 Mutation (Exploitation & Diversity Phase)

$$X_i^{t+1} = X_i^t + r_2 \times (X_j^t - X_k^t) \tag{5.15}$$

5.2.6 Adaptive Mutation Rate

$$M(t) = M_0 \times \left(1 - \frac{t}{T}\right) \tag{5.16}$$

5.2.7 Offspring Update with Mutation

$$X_i^{t+1} = X_i^t + M(t) \times rand(-1, 1) \times (X_{best}^t - X_i^t) \tag{5.17}$$

5.2.8 Boundary Control

$$x_{ij} = \begin{cases} x_j^{\min}, & x_{ij} < x_j^{\min} \\ x_j^{\max}, & x_{ij} > x_j^{\max} \end{cases} \tag{5.18}$$

5.3 Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO) is a population-based algorithm inspired by bird flocking. A "swarm" of candidate solutions (particles) navigates the search space, updating their positions by mimicking both their individual best-known positions and the entire swarm's global best.

5.3.1 Population Initialization

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{id}) \tag{5.19}$$

$$x_{ij} = x_j^{\min} + rand(0,1) \times (x_j^{\max} - x_j^{\min}) \tag{5.20}$$

5.3.2 Velocity Initialization

$$V_i = (v_{i1}, v_{i2}, v_{i3}, \ldots, v_{id}) \tag{5.21}$$

$$v_{ij} = v_j^{\min} + rand(0,1) \times (v_j^{\max} - v_j^{\min}) \tag{5.22}$$

5.3.3 Fitness Evaluation

$$f_i = f(X_i) \tag{5.23}$$

5.3.4 Personal Best (pbest) Update

$$P_{best,i}^{t+1} = \begin{cases} X_i^{t+1}, & \text{if } f(X_i^{t+1}) < f(P_{best,i}^t) \\ P_{best,i}^t, & \text{otherwise} \end{cases} \tag{5.24}$$

5.3.5 Global Best (gbest) Selection

$$G_{best}^t = \arg\min_{P_{best,i}^t} f(P_{best,i}^t), \quad \forall i \in \{1, 2, \ldots, N\} \tag{5.25}$$

5.3.6 Inertia Weight Calculation

$$w(t) = w_{\max} - \left(\frac{w_{\max} - w_{\min}}{T} \right) \times t \tag{5.26}$$

5.3.7 Velocity Update (Exploration & Exploitation)

$$V_i^{t+1} = w(t) \times V_i^t + c_1 \times r_1 \times (P_{best,i}^t - X_i^t) + c_2 \times r_2 \times (G_{best}^t - X_i^t) \tag{5.27}$$

5.3.8 Position Update

$$X_i^{t+1} = X_i^t + V_i^{t+1} \tag{5.28}$$

5.3.9 Boundary Control

$$x_{ij} = \begin{cases} x_j^{\min}, & x_{ij} < x_j^{\min} \\ x_j^{\max}, & x_{ij} > x_j^{\max} \end{cases} \tag{5.29}$$

5.4 Gorilla Troop Optimization (GTO)

Gorilla Troop Optimization (GTO) is a swarm-intelligence metaheuristic inspired by the social structures of gorillas. Candidate solutions act as troop members navigating a search space. Exploration models migration to unknown locations, while exploitation simulates following a dominant silver back leader and competing for adult females to find global optima.

5.4.1 Population Initialization

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{id}) \tag{5.30}$$

$$x_{ij} = x_j^{\min} + rand(0,1) \times (x_j^{\max} - x_j^{\min}) \tag{5.31}$$

5.4.2 Fitness Evaluation

$$f_i = f(X_i) \tag{5.32}$$

5.4.3 Internal Parameter Calculations

$$F = \cos(2 \times r_4) + 1 \tag{5.33}$$

$$C = F \times \left(1 - \frac{t}{T}\right) \tag{5.34}$$

$$L = C \times l, \quad H = Z \times X_i^t, \quad Z = [-C, C] \tag{5.35}$$

5.4.4 Exploration Phase (Migration Behavior)

The candidate position vector GX_i^{t+1} is generated based on a parameter p to choose between migrating to an unknown location, moving toward other gorillas, or moving

toward a known location:

$$GX_i^{t+1} = \begin{cases} (x_j^{\max} - x_j^{\min}) \times r_1 + x_j^{\min}, & \text{if } rand < p \\ (r_2 - C) \times X_r^t + L \times H, & \text{if } rand \geq 0.5 \\ X_i^t - L \times (L \times (X_i^t - GX_r^t) + r_3 \times (X_i^t - GX_r^t)), & \text{if } rand < 0.5 \end{cases} \tag{5.36}$$

(where X_r^t and GX_r^t are randomly selected gorilla individuals from the population).

5.4.5 Exploitation Phase (Silverback Leadership & Competition)

The transition from exploration to exploitation is controlled by the variable C. If $C \geq W$ (where W is a predefined parameter), the troop follows the silverback; otherwise, they compete for adult females.

5.4.5.1 Following the Silverback

$$GX_i^{t+1} = L \times M \times (X_i^t - X_{silverback}) + X_i^t \tag{5.37}$$

$$M = \left(\frac{1}{N} \sum_{i=1}^{N} |GX_i^t| \right)^{1/C^2} \tag{5.38}$$

5.4.5.2 Competition for Adult Females

$$GX_i^{t+1} = X_{silverback} - (X_{silverback} \times Q - X_i^t \times Q) \times E \tag{5.39}$$

$$Q = 2 \times r_5 - 1, \quad E = N_1 \times \mathcal{N}(0, 1) \tag{5.40}$$

5.4.6 Position Update

A greedy selection mechanism updates the gorilla's actual position only if the newly evaluated candidate position yields a better fitness score.

$$X_i^{t+1} = \begin{cases} GX_i^{t+1}, & \text{if } f(GX_i^{t+1}) < f(X_i^t) \\ X_i^t, & \text{otherwise} \end{cases} \tag{5.41}$$

5.4.7 Boundary Control

$$x_{ij} = \begin{cases} x_j^{\min}, & x_{ij} < x_j^{\min} \\ x_j^{\max}, & x_{ij} > x_j^{\max} \end{cases} \tag{5.42}$$

5.5 Billiards Optimization (BO)

Billiards Optimization (BO) mimics the impact mechanics of billiard balls, using velocity and angle changes after collisions to find optimal solutions.

5.5.1 Population Initialization

The population consists of $2N$ individuals representing billiard balls distributed in an M-dimensional search space.

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{iM}) \tag{5.43}$$

$$x_{im} = x_m^{\min} + rand(0,1) \times (x_m^{\max} - x_m^{\min}) \tag{5.44}$$

(where $i = 1, 2, \ldots, 2N$ and $m = 1, 2, \ldots, M$).

5.5.2 Fitness Evaluation

Each ball's position is evaluated using the objective function to establish its current fitness.

$$f_i = f(X_i) \tag{5.45}$$

5.5.3 Determination of Pockets (Target Solutions)

The 8 top-performing balls with the best fitness values are designated as "pockets" (P_k). These act as local target attractors and a memory repository.

$$P_k = \text{Top}_k \arg\min_{X_i} f(X_i), \quad \forall k \in \{1, 2, \ldots, 8\} \tag{5.46}$$

5.5.4 Ball Stratification & Grouping

The remaining population is sorted by fitness and split evenly into two distinct categories:

- **Regular Balls:** The superior half of the remaining population ($n = 1, 2, \ldots, N$).
- **Cue Balls:** The inferior half of the population ($n = N + 1, \ldots, 2N$).

Each cue ball is paired directly with a regular ball of the corresponding rank.

5.5.5 Pocket Assignment Mechanism

A roulette-wheel selection scheme assigns a target pocket P_{target} to each regular ball based on pocket fitness weights, balancing global and localized search:

$$W_k = \frac{1/f(P_k)}{\sum_{j=1}^{8}(1/f(P_j))} \tag{5.47}$$

5.5.6 Kinematic Position Update (Collision Mechanics)

The trajectory update relies on vector algebra derived from the conservation of momentum during elastic collisions. The position of the regular ball updates relative to its paired cue ball and the selected pocket target:

$$X_n^{t+1} = X_n^t + r_1 \times (P_{target}^t - X_n^t) + r_2 \times (X_{cue,n}^t - X_n^t) \tag{5.48}$$

(where r_1 and r_2 are random vectors uniformly distributed within $[0, 1]$).

5.5.7 Adaptive Search Space Narrowing

To transition systematically from exploration to exploitation, the upper and lower search boundaries are periodically squeezed around the historical best coordinate c_B using standard deviation values (sd_{values}):

$$x_m^{\min}(t+1) = c_{B,m} - sd_{values,m} \tag{5.49}$$

$$x_m^{\max}(t+1) = c_{B,m} + sd_{values,m} \tag{5.50}$$

5.5.8 Boundary Control

$$x_{im} = \begin{cases} x_m^{\min}, & x_{im} < x_m^{\min} \\ x_m^{\max}, & x_{im} > x_m^{\max} \end{cases} \tag{5.51}$$

5.6 Improved Social Group Optimization (ISGO)

Improved Social Group Optimization (ISGO) is a human-behavior-based algorithm simulating social interactions. It enhances standard SGO's improving and acquiring phases by introducing a self-awareness probability, allowing individuals to learn more effectively from the best-knowledgeable group member.

5.6.1 Population Initialization

The population consists of N persons representing individual solution candidates in a D-dimensional search space.

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{iD}) \tag{5.52}$$

$$x_{ij} = x_j^{\min} + rand(0,1) \times (x_j^{\max} - x_j^{\min}) \tag{5.53}$$

(where $i = 1, 2, \ldots, N$ and $j = 1, 2, \ldots, D$).

5.6.2 Fitness Evaluation

Each person's level of knowledge is quantified using the objective function to establish their current fitness value.

$$f_i = f(X_i) \tag{5.54}$$

5.6.3 Identification of the Group Best

The person possessing the highest level of knowledge is designated as the global leader (X_{gbest}).

$$X_{gbest} = \arg\min_{X_i} f(X_i), \quad \forall i \in \{1, 2, \ldots, N\} \tag{5.55}$$

5.6.4 Improving Phase

In this phase, every individual improves their individual knowledge base under the direct influence (X_{gbest}).

$$X_{i,\text{new}} = c \times X_{i,\text{old}} + r \times (X_{gbest} - X_{i,\text{old}}) \tag{5.56}$$

(where $c \in (0, 1)$ is the self-introspection factor, and $r \sim U(0, 1)$).

5.6.5 Acquiring Phase with Self-Awareness Probability

In standard SGO, an individual interacts randomly with another person (X_r). In ISGO a Self-Awareness Probability (SAP) parameter is integrated to decide whether the person updates knowledge via a random peer to the group leader.

For each individual X_i, a random person X_r ($r \neq i$) is chosen.

5.6.5.1 Case 1: When $rand(0, 1) > SAP$

The standard interaction mechanism based on relative knowledge dominance occurs:

If $f(X_i) < f(X_r)$ (i.e., X_i is better than X_r):

$$X_{i,\text{new}} = X_{i,\text{old}} + r_1 \times (X_i - X_r) + r_2 \times (X_{gbest} - X_{i,\text{old}}) \tag{5.57}$$

If $f(X_i) \geq f(X_r)$ (i.e., X_r is better than X_i):

$$X_{i,\text{new}} = X_{i,\text{old}} + r_1 \times (X_r - X_i) + r_2 \times (X_{gbest} - X_{i,\text{old}}) \tag{5.58}$$

5.6.5.2 Case 2: When $rand(0, 1) \leq SAP$

The self-awareness learning component directly updates the position to fine-tune around the best knowledgeable person:

$$X_{i,\text{new}} = X_{gbest} + r_3 \times (X_{gbest} - X_{i,\text{old}}) \tag{5.59}$$

(where $r_1, r_2, r_3 \sim U(0, 1)$ are independent random vector paths).

5.6.6 Selection Operator

A greedy selection routine updates the group member's true state only if the newly acquired knowledge results in an improved fitness.

$$X_i^{t+1} = \begin{cases} X_{i,\text{new}}, & \text{if } f(X_{i,\text{new}}) < f(X_i^t) \\ X_i^t, & \text{otherwise} \end{cases} \tag{5.60}$$

5.6.7 Boundary Control

$$x_{ij} = \begin{cases} x_j^{\min}, & x_{ij} < x_j^{\min} \\ x_j^{\max}, & x_{ij} > x_j^{\max} \end{cases} \tag{5.61}$$

5.7 Energy Optimization (EO)

Energy Optimization (EO) is a general application field (e.g., minimizing power consumption in smart grids or cloud networks). The physical concept underlying algorithms like Simulated Annealing, which mimic molecules cooling to reach their lowest energy state.

5.7.1 Decision Variables

Let X_i represent the i-th candidate solution vector containing the operational parameters or states of energy-consuming assets in a d-dimensional space.

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{id}) \tag{5.62}$$

5.7.2 Objective Function (Total Energy Minimization)

The primary goal is to minimize the total energy consumption (E_{total}) over a specific time horizon T, which is computed as the sum of power ratings (P_j) multiplied by operational durations or status variables (x_{ij}):

$$f(X_i) = \min \sum_{t=1}^{T} \sum_{j=1}^{d} P_j \times x_{ij}(t) \times \Delta t \tag{5.63}$$

(where $x_{ij}(t) \in [0, 1]$ can represent binary ON/OFF states or continuous throttling levels).

5.7.3 System Constraints

5.7.3.1 Demand-Supply Balancing Constraint

The total optimized energy output must satisfy the minimum required load demand (D_t) at any time step t:

$$\sum_{j=1}^{d} E_j(t) \geq D_t, \quad \forall t \in \{1, 2, \ldots, T\} \tag{5.64}$$

5.7.3.2 Operational Threshold Constraints

Each asset must operate within its safe minimum and maximum capacity limitations:

$$E_j^{\min} \leq E_j(t) \leq E_j^{\max} \tag{5.65}$$

5.7.4 Optimization Cost Function (Weighted Evaluation)

In practical scenarios, energy minimization is balanced against penalty costs (such as user discomfort, peak-demand charges, or delay penalties) using weighting factors (α and β):

$$F(X_i) = \alpha \times f(X_i) + \beta \times \text{Penalty}(X_i) \tag{5.66}$$

(where $\alpha + \beta = 1$).

5.7.5 Boundary Control

Any decision variable vector updating via a metaheuristic method (such as GA or PSO) is strictly mapped back to its physical bounds:

$$x_{ij} = \begin{cases} x_j^{\min}, & x_{ij} < x_j^{\min} \\ x_j^{\max}, & x_{ij} > x_j^{\max} \end{cases} \tag{5.67}$$

5.8 Hippo Optimization (HO)

Hippo Optimization (HO) algorithm is a nature-inspired metaheuristic that models the survival patterns of hippos. It utilizes a trinary-phase model—river/pond position updates, defensive mechanics against predators, and evasion strategies—to balance global exploration and local exploitation.

5.8.1 Population Initialization

The initial population consists of N hippopotamuses randomly distributed as candidate solution vectors within a m-dimensional space.

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{im}) \tag{5.68}$$

$$x_{ij} = x_j^{\min} + rand(0,1) \times (x_j^{\max} - x_j^{\min}) \tag{5.69}$$

(where $i = 1, 2, \ldots, N$ and $j = 1, 2, \ldots, m$).

5.8.2 Fitness Evaluation

The fitness of each individual hippopotamus is determined based on the objective function.

$$f_i = f(X_i) \tag{5.70}$$

5.8.3 Phase 1: Position Updating in Rivers or Ponds (Exploration Phase)

This phase simulates the herd's behavior inside a water body, led by a dominant male (D_h). The behavior splits between the adult movement and maternal protection of calves based on a probability threshold T:

5.8.3.1 Dominant Male Alignment

$$X_{i,\text{new1}} = X_i^t + y_1 \times (D_h^t - I_1 \times X_i^t) \tag{5.71}$$

5.8.3.2 Maternal Protection and Calf Proximity

$$X_{i,\text{new2}} = \begin{cases} X_i^t + h_1 \times (D_h^t - I_2 \times MG_i^t), & \text{if } T > 0.6 \\ X_i^t + h_2 \times (X_j^t - X_k^t), & \text{otherwise} \end{cases} \tag{5.72}$$

(where y_1, h_1, h_2 are random numbers/vectors in $[0, 1]$, I_1, I_2 are random integers selected from $\{1, 2\}$, and MG_i^t represents the mean position of the group).

5.8.4 Phase 2: Defensive Strategies against Predators (Exploration/Exploitation Boundary)

Hippos exhibit aggressive defensive behaviors when predators (lions or hyenas) approach. The position of a virtual predator is generated randomly:

$$Predator_j = x_j^{\min} + rand(0, 1) \times (x_j^{\max} - x_j^{\min}) \tag{5.73}$$

The distance vector between the predator and the hippo determines the defensive stance:

$$\vec{D}_i = Predator_j - X_i^t \tag{5.74}$$

The updated defensive position forces the hippo to charge or group defensively:

$$X_{i,\text{def}} = X_i^t + r_1 \times \vec{D}_i \tag{5.75}$$

(where r_1 is a uniform random parameter adjusting the reaction intensity).

5.8.5 Phase 3: Evasion Methods (Exploitation Phase)

When fleeing a danger zone, hippos seek optimal safety within localized constraints. The search boundaries are dynamically restricted to create a local exploitative range:

$$LB_j^{\text{local}} = \frac{x_j^{\min}}{t}, \quad UB_j^{\text{local}} = \frac{x_j^{\max}}{t} \tag{5.76}$$

The final position update refines the solution agent within this highly focused range:

$$X_i^{t+1} = X_i^t + r_2 \times LB_j^{\text{local}} + r_3 \times (UB_j^{\text{local}} - LB_j^{\text{local}}) \tag{5.77}$$

(where r_2, r_3 act as stochastic local weights, and t tracks the current iteration).

5.8.6 Boundary Control

$$x_{ij} = \begin{cases} x_j^{\min}, & x_{ij} < x_j^{\min} \\ x_j^{\max}, & x_{ij} > x_j^{\max} \end{cases} \tag{5.78}$$

5.9 Meta-OOD

Meta-OOD primarily refers to the application of meta-learning frameworks to Out-of-Distribution (OOD) detection. The core objective is to handle "open-world" scenarios where a deployed model encounters data distributions completely different from its original training data.

5.9.1 Problem Setup and Distribution Domain

Let X_i represent a d-dimensional test sample vector. The input space consists of an In-Distribution (ID) domain $\mathcal{D}_{ID}$ and an unknown, shifting Out-of-Distribution (OOD) domain $\mathcal{D}_{OOD}$ whose semantic spaces are disjoint.

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{id}) \tag{5.79}$$

5.9.2 Base OOD Scoring Mechanism

A baseline model evaluates an input vector using a chosen post-hoc scoring function $s(X_i; f)$ (such as energy score or softmax entropy) to produce a scalar metric:

$$s_i = s(X_i; f) \tag{5.80}$$

The standard detection routing relies on a hard threshold parameter γ:

$$G_\gamma(X_i) = \begin{cases} \text{ID}, & \text{if } s(X_i; f) \geq \gamma \\ \text{OOD}, & \text{if } s(X_i; f) < \gamma \end{cases} \tag{5.81}$$

5.9.3 MetaOOD Selection Predictor (Offline Phase)

Instead of manually tuning γ or selecting a base detector blindly, a meta-predictor (e.g., an optimized gradient-boosted tree framework) maps meta-features of a target dataset to expected detector performance.

Let $\mathcal{E}_D$ represent language model dataset text embeddings, and $\mathcal{E}_M$ represent model property embeddings. The performance estimator $\hat{P}$ is modeled as:

$$\hat{P}(D, M) = \mathcal{M}_{predictor}\left([\mathcal{E}_D \parallel \mathcal{E}_M]\right) \tag{5.82}$$

(where $\parallel$ denotes vector concatenation).

The chosen OOD model configuration M^* selected for execution is solved via argmax:

$$M^* = \arg\max_{M_k \in \mathcal{M}} \hat{P}(D_{target}, M_k) \tag{5.83}$$

5.9.4 Meta OOD Adaptation Loop (Online Phase)

In continuously adaptive meta-OOD variations, the objective turns to optimizing a lightweight adaptation layer mapping parameterized by θ. Under task distribution updates, a localized loss calculation minimizing uncertainty (entropy H) optimizes the model parameters over few-shot support task sets $\mathcal{T}_t$:

$$\mathcal{L}_{\text{meta}}(\theta) = \sum_{\mathcal{T}_t} H\left(G_\gamma(X_i; \theta)\right) \tag{5.84}$$

The optimization engine parameters update across continuous meta-steps scaling by a task learning rate step α:

$$\theta^{t+1} = \theta^t - \alpha \nabla_\theta \mathcal{L}_{\text{meta}}(\theta^t) \tag{5.85}$$

5.9.5 Boundary Control

To prevent divergence during gradient-based meta-adaptation steps or meta-feature scaling, parameters are explicitly bound within target structural constraints:

$$\theta_j = \begin{cases} \theta_j^{\min}, & \theta_j < \theta_j^{\min} \\ \theta_j^{\max}, & \theta_j > \theta_j^{\max} \end{cases} \tag{5.86}$$

5.10 Sequential Pattern Mining Optimization (SPM)

Sequential Pattern Mining Optimization (SPM) speeds up discovery of frequent subsequences in data by pruning unpromising search paths using constraints.

5.10.1 Sequence Database Definition

Let $I = \{i_1, i_2, \ldots, i_m\}$ be a set of items. An itemset $S_k \subseteq I$ is a collection of items. A sequence α is an ordered list of itemsets:

$$\alpha = \langle S_1, S_2, S_3, \ldots, S_d \rangle \tag{5.87}$$

A sequence database D is a collection of tuples (sid, α_i), where sid is a unique sequence identifier and α_i is a sequence.

5.10.2 Sequence Support Metric

The absolute support of a sequence α in database D corresponds to the total number of sequences that contain α as a subsequence:

$$\text{Sup}(\alpha) = |\{(sid, \alpha_i) \in D \mid \alpha \sqsubseteq \alpha_i\}| \tag{5.88}$$

(where $\sqsubseteq$ denotes the subsequence containment relation).

The relative support calculation normalizes this count by the total database size N:

$$f_i = \frac{\text{Sup}(\alpha_i)}{N} \tag{5.89}$$

5.10.3 Anti-Monotonicity Principle (Apriori Pruning)

To optimize search space navigation, the downward-closure property ensures that if a sequence α is infrequent relative to a minimum support threshold ($minsup$), all its super-sequences (β) are pruned automatically:

$$\forall \alpha, \beta : (\alpha \sqsubseteq \beta) \wedge (\text{Sup}(\alpha) < minsup) \implies \text{Sup}(\beta) < minsup \tag{5.90}$$

5.10.4 Sequence-Extension Matrix (PrefixSpan Projection)

In projection-based pattern growth optimization, a sequence is expanded dynamically. Given a prefix sequence α, the database is projected into a reduced search subspace $D|_\alpha$:

$$D|_\alpha = \{\alpha_i' \mid \exists(sid, \alpha_i) \in D \text{ s.t. } \alpha_i = \alpha \cdot \alpha_i'\} \tag{5.91}$$

The candidate position vector for the next expansion step updates exclusively within this restricted sub-context:

$$\alpha^{t+1} = \alpha^t \cdot \text{item}_j \tag{5.92}$$

5.10.5 Constraint-Based Boundary Bounds

When processing time-interval or gap constraints, sequences violate optimization conditions if the time difference ($\Delta\tau$) between successive itemsets exceeds predefined minimum ($\min_{gap}$) or maximum ($\max_{gap}$) operational ranges:

$$\Delta\tau = \tau(S_{k+1}) - \tau(S_k) \tag{5.93}$$

The validation routine filters invalid sequence expansion paths using a hard step constraint:

$$\Delta\tau = \begin{cases} \min_{gap}, & \Delta\tau < \min_{gap} \\ \max_{gap}, & \Delta\tau > \max_{gap} \end{cases} \tag{5.94}$$

General Trend (GT) in data analytics or optimization tracks the long-term upward, downward, or stationary direction of variables over time, isolating sustained patterns from noise or short-term seasonal fluctuations.

5.11 Mathematical Expression

5.11.1 Time Series Representation

Let X_t represent a d-dimensional observation vector recorded at a specific sequential time index t within a finite historical window T.

$$X_t = (x_{t1}, x_{t2}, x_{t3}, \ldots, x_{td}) \tag{5.95}$$

5.11.2 Additive Decomposition Model

The raw data signal is broken down into three independent, unobserved components: the general trend component (T_t), the cyclical variation (S_t) and an irregular random noise residual (ϵ_t):

$$X_t = T_t + S_t + \epsilon_t \tag{5.96}$$

5.11.3 Trend Extraction (Moving Average Filter)

To isolate the general trend and smooth out localized high frequency noise, a centered symmetric moving average filter of window span k is applied to the time series:

$$T_t = \frac{1}{2k+1} \sum_{i=-k}^{k} X_{t+i} \tag{5.97}$$

5.11.4 Linear Trend Optimization (Objective Function)

When modeling the general trend via parametric regression, the optimal trend trajectory parameters—slope vector m and intercept vector b—are solved by minimizing the sum of

squared errors (f_i):

$$f_i = \min_{m,b} \sum_{t=1}^{T} |X_t - (m \times t + b)|^2 \tag{5.98}$$

5.11.5 Stochastic Trend Velocity Update

In adaptive filtering and predictive optimization networks, the trend directional state update vectors scale dynamically over iterations based on the structural velocity change vector (V_t):

$$T_t^{t+1} = T_t^t + r_1 \times V_t^t + r_2 \times (X_{\text{target}}^t - T_t^t) \tag{5.99}$$

(where r_1, r_2 are random coefficient matrices uniformly distributed within $[0, 1]$).

5.11.6 Boundary Control

To maintain structural validity and prevent trend projections from drifting into unrealistic physical values, values are bounded within system parameters:

$$T_t = \begin{cases} T^{\min}, & T_t < T^{\min} \\ T^{\max}, & T_t > T^{\max} \end{cases} \tag{5.100}$$

5.12 Computational Complexity

The computational complexity of the proposed Improved Fungal Growth Optimization (IFGO) algorithm depends on the population size P, number of iterations T, dataset size N, number of clusters K, and feature dimension d.

In each iteration, the dominant computational cost arises from the cluster assignment process, where each data point is assigned to its nearest centroid. This requires computing distances between N data points and K centroids, resulting in a complexity of $\mathcal{O}(NK)$ per candidate solution. Considering P candidate solutions over T iterations the overall time complexity of the algorithm can be expressed as:

$$\mathcal{O}(PTNK) \tag{5.101}$$

Additional operations such as Lévy flight-based mutation and opposition-based learning introduce an extra computational cost of $\mathcal{O}(P)$ per iteration. However, this cost is relatively negligible compared to the dominant clustering computation.

In comparison, traditional clustering algorithms such as K-Means and Fuzzy C-Means (FCM) exhibit a time complexity of $\mathcal{O}(NKT)$. The higher computational cost of IFGO is attributed to its population-based optimization framework. Despite this, IFGO provides improved global search capability, better avoidance of local optima, and enhanced clustering performance.

The space complexity of the proposed algorithm is determined by the storage of candidate solutions and is given by (where d denotes the dimensionality of the dataset):

$$\mathcal{O}(PKd) \tag{5.102}$$

Chapter 6

Results and Discussion

The results are presented here. The following cluster validity indices and statistical analyses are used to compare the quality of all formed clusters.

Silhouette Score [25] measures the degree to which each data point fits within its assigned cluster. High value represents better separation and cohesion between the formed clusters.

Davies-Bouldin Index [26] evaluates the mean similarity between clusters. Less value represents clustering with more distinct groups.

Dunn Index [27] evaluates cluster separation and compactness. High value indicates well-separated clusters.

Xie-Beni Index [28] assesses compactness and separation. Low value represents well-defined clusters.

Caliński-Harabasz Index [29] is used to determine the optimal number of clusters. Higher value shows better results.

TABLE 6.1: Description of the Datasets used in the Experimental Study

Dataset	**Domain**	**Sample**	**Features**
Iris Dataset	Botany	150	6
Drug Dataset	Healthcare / Pharmacology	1,000	9
Bank Dataset	Finance	2,512	16
Shopping Dataset	E-Commerce	3,900	18
Wave Dataset	Signal Processing	12,762	3
Placement Dataset	Education	10,000	10
Delivery Dataset	Supply Chain	25,000	15

Table 6.1 summarizes the domain, number of samples, and features count for each dataset used in the study. These datasets are available in the public domain, downloaded from the website Kaggle [30].

6.1 Comparison of Clustering Validity Indices

TABLE 6.2: Comparison of Clustering Validity Indices on the Bank Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
IFGO+K-Means	1.000000000	0.000000000	1.000000000	0.030107357	0.010134850
IFGO+FCM	0.986653205	0.001939714	0.991290384	0.030107357	0.012817260
FGO+K-Means	0.891107530	0.004169055	0.786877028	0.022056461	0.000330080
FGO+FCM	0.707623194	0.031248446	0.530932021	0.000103282	0.054457788
K-Means	0.890768774	0.004166439	0.786857872	0.000000000	0.000386221
FCM	0.707210793	0.032016781	0.531045550	0.000000000	0.054456663
GA+K-Means	0.891251131	0.004177170	0.786875215	0.022056461	0.000236122
GA+FCM	0.707221840	0.031673578	0.530827932	0.000103282	0.054368291
PSO+K-Means	0.891107530	0.004169055	0.786877028	0.022056461	0.000330080
PSO+FCM	0.707663700	0.031238108	0.531002131	0.000103282	0.054389852
GTO+K-Means	0.891107530	0.004169055	0.786877028	0.022056461	0.000330080
GTO+FCM	0.707446454	0.031777709	0.530749524	0.000103282	0.054523872
BO+K-Means	0.891107530	0.004169055	0.786877028	0.022056461	0.000330080
BO+FCM	0.707221840	0.031499043	0.530827932	0.000103282	0.054456663
ISGO+K-Means	0.891107530	0.004169055	0.786877028	0.022056461	0.000330080
ISGO+FCM	0.707446454	0.031774415	0.530749524	0.000103282	0.054523872
EO+K-Means	0.891107530	0.004169055	0.786877028	0.022056461	0.000330080
EO+FCM	0.707446454	0.031766961	0.530749524	0.000103282	0.054523872
HO+K-Means	0.891107530	0.004169055	0.786877028	0.022056461	0.000330080
HO+FCM	0.707914083	0.031240313	0.530980421	0.000103282	0.054371759
MetaOOD+K-Means	0.843085601	0.014111322	0.094635198	1.000000000	0.098039032
MetaOOD+FCM	0.000000000	1.000000000	0.000000000	0.749621300	0.000000000
SPM+K-Means	0.220059346	0.050251347	0.177555310	0.742230893	0.291093156
SPM+FCM	0.219322920	0.127635677	0.176553070	0.742230893	0.484891172
GT+K-Means	0.273826075	0.027220542	0.212546416	0.911314987	0.151121088
GT+FCM	0.182473434	0.535579998	0.163395701	0.749368924	1.000000000

Table 6.2 showcases the comparison of clustering validity indices on the Bank Dataset. The IFGO integrated with K-Means outperforms other K-Means based hybrid algorithms, achieving the highest Silhouette Score, CH-Score and Dunn-Index, as well as the lowest XB-Index. These results suggest that the IFGO+K-Means approach provides superior cluster separation and compactness for this specific dataset. Similarly, the IFGO+FCM configuration demonstrates superior performance among the fuzzy-logic-based approaches.

TABLE 6.3: Comparison of Clustering Validity Indices on the Delivery Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
IFGO+K-Means	0.974792717	0.000000000	0.964312090	0.000885837	0.250907648
IFGO+FCM	0.611623247	0.000247924	0.834014739	0.000000000	0.341733177
FGO+K-Means	1.000000000	0.000203678	1.000000000	0.001461321	0.247151453
FGO+FCM	0.622477393	0.000517663	0.852318754	0.000579456	0.336878847
K-Means	0.999891450	0.000203610	0.999999740	0.001465785	0.247176952
FCM	0.622798324	0.000521447	0.852365002	0.000572908	0.336882359
GA+K-Means	1.000000000	0.000203678	1.000000000	0.001461321	0.000000000
GA+FCM	0.622515150	0.000517676	0.852319310	0.000579456	0.247151453
PSO+K-Means	1.000000000	0.000203678	1.000000000	0.001461321	0.247151453
PSO+FCM	0.622552906	0.000517654	0.852307786	0.000579456	0.336914577
GTO+K-Means	1.000000000	0.000203678	1.000000000	0.001461321	0.247151453
GTO+FCM	0.622472674	0.000517578	0.852311047	0.000579456	0.336910607
BO+K-Means	1.000000000	0.000203678	1.000000000	0.001461321	0.247151453
BO+FCM	0.622477393	0.000517668	0.852318754	0.000579456	0.336878847
ISGO+K-Means	1.000000000	0.000203678	1.000000000	0.001461321	0.247151453
ISGO+FCM	0.622472674	0.000517693	0.852311047	0.000579456	0.336910607
EO+K-Means	1.000000000	0.000203678	1.000000000	0.001461321	0.247151453
EO+FCM	0.622548187	0.000517657	0.852303556	0.000579456	0.336924196
HO+K-Means	1.000000000	0.000203678	1.000000000	0.001461321	0.247151453
HO+FCM	0.622444356	0.000517681	0.852311857	0.000579456	0.336874114
MetaOOD+K-Means	0.498201846	0.000638484	0.417178589	0.042865520	0.367527253
MetaOOD+FCM	0.000000000	1.000000000	0.000000000	0.034826054	0.188650854
SPM+K-Means	0.330185384	0.001854738	0.288239524	0.029948564	0.567398536
SPM+FCM	0.297148440	0.003420898	0.285230202	0.018103062	0.727478789
GT+K-Means	0.406642314	0.001572972	0.364549684	1.000000000	0.512277426
GT+FCM	0.175383701	0.006845344	0.272505177	0.029115212	1.000000000

Table 6.3 signifies that for the Delivery Dataset, K-Means based hybrid algorithms generally demonstrate superior performance compared to FCM based approaches across most validity indices. While standard K-Means and traditional metaheuristic hybrids provide highly competitive Silhouette and CH-Scores, the IFGO+K-Means algorithm achieves the most optimal XB-Index. This indicates that the proposed IFGO optimization effectively enhances cluster compactness and separation beyond the capabilities of standard initialization, even in cases where the dataset strongly favors a K-Means centroid structure.

TABLE 6.4: Comparison of Clustering Validity Indices on the Drug Dataset

Algorithm	**S-Score**	**XB-Index**	**CH-Index**	**Dunn-Index**	**DB-Index**
IFGO+K-Means	0.988397509	0.000000000	1.000000000	0.003499855	0.001267777
IFGO+FCM	1.000000000	0.000343581	0.993971538	0.030106499	0.000000000
FGO+K-Means	0.784729374	0.001084068	0.653942951	0.069255744	0.057096458
FGO+FCM	0.733616102	0.001553534	0.611286936	0.000019980	0.051049725
K-Means	0.784678328	0.001084074	0.653942987	0.069430569	0.057084277
FCM	0.729095765	0.001563302	0.609158599	0.000000000	0.050831556
GA+K-Means	0.784729374	0.001084068	0.653942951	0.069255744	0.057096458
GA+FCM	0.733616102	0.001553542	0.611286936	0.000019980	0.051049725
PSO+K-Means	0.784729374	0.001084068	0.653942951	0.069255744	0.057096458
PSO+FCM	0.733616102	0.001553534	0.611286936	0.000019980	0.051049725
GTO+K-Means	0.784729374	0.001084068	0.653942951	0.069255744	0.057096458
GTO+FCM	0.733616102	0.001553534	0.611286936	0.000019980	0.051049725
BO+K-Means	0.784729374	0.001084068	0.653942951	0.069255744	0.057096458
BO+FCM	0.733616102	0.001550793	0.611286936	0.000019980	0.051049725
ISGO+K-Means	0.784729374	0.001084068	0.653942951	0.069255744	0.057096458
ISGO+FCM	0.733616102	0.001553542	0.611286936	0.000019980	0.051049725
EO+K-Means	0.784729374	0.001084068	0.653942951	0.069255744	0.057096458
EO+FCM	0.733616102	0.001553534	0.611286936	0.000019980	0.051049725
HO+K-Means	0.784729374	0.001084068	0.653942951	0.069255744	0.057096458
HO+FCM	0.733616102	0.001553534	0.611286936	0.000019980	0.051049725
MetaOOD+K-Means	0.058701993	0.003703080	0.239868926	0.948551449	0.343328967
MetaOOD+FCM	0.000000000	1.000000000	0.000000000	0.949255744	0.679297675
SPM+K-Means	0.086493274	0.007074166	0.253848536	0.964035964	0.494179242
SPM+FCM	0.075149894	0.019725784	0.251561498	0.950049950	0.938609650
GT+K-Means	0.096135148	0.004663340	0.255113455	1.000000000	0.385934087
GT+FCM	0.067209528	0.016115858	0.241373064	0.908091908	1.000000000

Table 6.4 demonstrates that hybrid algorithms integrated with the IFGO optimization significantly outperform all other evaluated models on the Drug Dataset. Notably, the IFGO+FCM configuration achieves the highest overall Silhouette Score and the most optimal DB-Index, indicating exceptionally well-defined cluster boundaries. Simultaneously, IFGO+K-Means delivers superior results in cluster compactness and variance-based structure, producing the highest CH-Index and the lowest XB-Index. These results confirm that the IFGO algorithm substantially enhances clustering quality, providing a more robust separation between groups.

TABLE 6.5: Comparison of Clustering Validity Indices on the Iris Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
IFGO+K-Means	0.964053102	0.000000000	1.000000000	0.442841424	0.042782204
IFGO+FCM	0.964053102	0.006421797	1.000000000	0.442841424	0.042782204
FGO+K-Means	0.009206799	0.307947633	0.442969922	0.030260374	0.212257480
FGO+FCM	0.000269122	0.190536733	0.442236503	0.030260374	0.210983033
K-Means	0.008498584	0.308019201	0.442969843	0.030105777	0.212278202
FCM	0.000000000	0.190966140	0.442236512	0.030105777	0.210983033
GA+K-Means	0.009206799	0.307947633	0.442969922	0.030260374	0.212257480
GA+FCM	0.000269122	0.190966140	0.442236503	0.030260374	0.212257480
PSO+K-Means	0.009206799	0.307947633	0.442969922	0.030260374	0.212257480
PSO+FCM	0.000269122	0.190501745	0.442236503	0.030260374	0.210983033
GTO+K-Means	0.009206799	0.307947633	0.442969922	0.030260374	0.212257480
GTO+FCM	0.000269122	0.190544685	0.442236503	0.030260374	0.210983033
BO+K-Means	0.009206799	0.307947633	0.442969922	0.030260374	0.212257480
BO+FCM	0.000269122	0.190557408	0.442236503	0.030260374	0.210983033
ISGO+K-Means	0.009206799	0.307947633	0.442969922	0.030260374	0.212257480
ISGO+FCM	0.000269122	0.190660784	0.442236503	0.030260374	0.210983033
EO+K-Means	0.009206799	0.307947633	0.442969922	0.030260374	0.212257480
EO+FCM	0.000269122	0.190681459	0.442236503	0.030260374	0.210983033
HO+K-Means	0.009206799	0.307947633	0.442969922	0.030260374	0.212257480
HO+FCM	0.000269122	0.190678278	0.442236503	0.030260374	0.210983033
MetaOOD+K-Means	0.103399433	0.990616670	0.000000000	0.000000000	0.000000000
MetaOOD+FCM	1.000000000	0.196532522	0.572460367	1.000000000	0.087164875
SPM+K-Means	0.947592068	0.902985914	0.564677020	0.266069976	0.933298795
SPM+FCM	0.947592068	1.000000000	0.564677020	0.266069976	1.000000000
GT+K-Means	0.678470255	0.560096785	0.519678875	0.030105777	0.561196736
GT+FCM	0.678470255	0.618782355	0.519678875	0.030105777	0.611190260

Table 6.5 displays that for the Iris Dataset, the hybrid algorithms based on the IFGO provide the overall highest Silhouette Score, CH-Score and Dunn-Index while achieving the lowest XB-Index. Furthermore it is observed that other traditional metaheuristic-based hybrid algorithms yield nearly identical clustering validity indices. This suggests that while standard metaheuristics converge to similar local optima, the IFGO successfully explores a more optimal solution space resulting in superior cluster separation and structural compactness.

TABLE 6.6: Comparison of Clustering Validity Indices on the Placement Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
IFGO+K-Means	0.746481298	0.000000000	0.570894675	0.224195740	0.000283744
IFGO+FCM	1.000000000	1.000000000	1.000000000	0.273421947	0.000000000
FGO+K-Means	0.777658031	0.000057454	0.692786841	0.000000000	0.003680945
FGO+FCM	0.131796793	0.018835026	0.353701739	0.264553995	0.052896153
K-Means	0.777065089	0.000057465	0.692781656	0.000352573	0.003680945
FCM	0.133411777	0.018982450	0.353816957	0.263745152	0.052909826
GA+K-Means	0.777658031	0.000057454	0.692786841	0.000000000	0.003680945
GA+FCM	0.133084099	0.018981591	0.353816997	0.264553995	0.052909632
PSO+K-Means	0.777658031	0.000057454	0.692786841	0.000000000	0.003680945
PSO+FCM	0.131796793	0.018830560	0.353701739	0.264553995	0.052896153
GTO+K-Means	0.777658031	0.000057454	0.692786841	0.000000000	0.003680945
GTO+FCM	0.133084099	0.018984871	0.353816997	0.264553995	0.052909632
BO+K-Means	0.777658031	0.000057454	0.692786841	0.000000000	0.003680945
BO+FCM	0.133084099	0.019003497	0.353816997	0.264553995	0.052909632
ISGO+K-Means	0.777658031	0.000057454	0.692786841	0.000000000	0.003680945
ISGO+FCM	0.133084099	0.018993962	0.353816997	0.264553995	0.052909632
EO+K-Means	0.777658031	0.000057454	0.692786841	0.000000000	0.003680945
EO+FCM	0.133084099	0.018984850	0.353816997	0.264553995	0.052909632
HO+K-Means	0.777658031	0.000057454	0.692786841	0.000000000	0.003680945
HO+FCM	0.133076297	0.019027447	0.353681053	0.264553995	0.052934654
MetaOOD+K-Means	0.617907179	0.000072813	0.335854404	1.000000000	0.013563937
MetaOOD+FCM	0.002090898	0.055677859	0.000000000	0.400937429	0.213388220
SPM+K-Means	0.555492313	0.000119958	0.329393164	0.852749030	0.027589480
SPM+FCM	0.090501557	0.002620443	0.142462369	0.852749030	0.302166303
GT+K-Means	0.549250826	0.000114136	0.320402270	0.852749030	0.024905221
GT+FCM	0.000000000	0.026505962	0.079692733	0.400626335	1.000000000

Table 6.6 illustrates a relatively low Silhouette Score across all evaluated hybrid algorithms, indicating that the clustering structure within the Placement Dataset is inherently weak or highly overlapped. Despite these structural challenges the IFGO provides a moderate enhancement to both K-Means and FCM performance, specifically achieving the highest Silhouette Score and CH-Score with the FCM variant. The exceptionally high XB-Index values observed across several models (including the fuzzy-based configurations), suggest a degree of instability or high sensitivity to centroid placement. These findings underscore the relative robustness of the IFGO optimization in extracting meaningful patterns where standard approaches struggle.

TABLE 6.7: Comparison of Clustering Validity Indices on the Shopping Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
IFGO+K-Means	1.000000000	0.000000000	1.000000000	0.014607249	0.013010725
IFGO+FCM	0.325903813	0.000666478	0.103649598	0.000986907	0.026706429
FGO+K-Means	0.983478705	0.000000017	0.239125623	0.042234924	0.000000000
FGO+FCM	0.895052277	0.017345670	0.212426334	0.026904839	0.007696377
K-Means	0.984991370	0.000000018	0.238841998	0.003328589	0.000671002
FCM	0.847796143	0.037981447	0.202196152	0.011863432	0.010245750
GA+K-Means	0.983478705	0.000000017	0.239125623	0.042234924	0.000000000
GA+FCM	0.897374042	0.019322058	0.210502345	0.043045734	0.009400580
PSO+K-Means	0.983478705	0.000000017	0.239125619	0.042234924	0.000000000
PSO+FCM	0.892402181	0.016155126	0.210685984	0.016336695	0.007456495
GTO+K-Means	0.983478705	0.000000017	0.239125623	0.042234924	0.000000000
GTO+FCM	0.896928451	0.016047316	0.209027053	0.034721752	0.008960860
BO+K-Means	0.983478705	0.000000017	0.239125623	0.042234924	0.000000000
BO+FCM	0.897151247	0.016692111	0.210992469	0.010686126	0.008876536
ISGO+K-Means	0.983478705	0.000000017	0.239125623	0.042234924	0.000000000
ISGO+FCM	0.903072921	0.017457910	0.210612960	0.000000000	0.007945757
EO+K-Means	0.983478705	0.000000017	0.239125623	0.042234924	0.000000000
EO+FCM	0.905957539	0.016924984	0.213316959	0.039837637	0.008233641
HO+K-Means	0.983478705	0.000000017	0.239125623	0.042234924	0.000000000
HO+FCM	0.904515230	0.015972493	0.212107672	0.018028602	0.008692613
MetaOOD+K-Means	0.426829594	0.000000192	0.068901413	0.830957384	0.174530966
MetaOOD+FCM	0.966464151	0.134779710	0.190496848	1.000000000	0.873205184
SPM+K-Means	0.152439141	0.000000669	0.060548163	0.759666338	0.595473326
SPM+FCM	0.123123921	0.000001830	0.057182312	0.768201181	1.000000000
GT+K-Means	0.263836974	0.000000216	0.055834508	0.633651884	0.267852669
GT+FCM	0.000000000	1.000000000	0.000000000	0.596645807	0.320321260

Table 6.7 denotes that for the Shopping Dataset, FCM-based hybrid algorithms demonstrate significant instability, as evidenced by extreme XB-index values and lower CH-scores, which indicate poor cluster compactness and substantial overlap. Among all evaluated approaches, IFGO+K-Means provides the most balanced performance by successfully optimizing the cluster structure. It achieves the highest CH-Score and the most optimal XB-Index effectively mitigating the structural weaknesses inherent in the dataset. These results suggest that the IFGO optimization offers a more stable and reliable configuration for centroid-based clustering.

TABLE 6.8: Comparison of Clustering Validity Indices on the Wave Dataset

Algorithm	**Silhouette**	**XB-Index**	**CH-Score**	**Dunn-Index**	**DB-Index**
IFGO+K-Means	1.000000000	0.000000000	1.000000000	0.015343092	0.000000000
IFGO+FCM	0.960123451	0.039640278	0.975182196	0.004955862	0.003284898
FGO+K-Means	0.547247236	0.442343813	0.146243505	0.005344237	0.363601128
FGO+FCM	0.539676953	0.359191975	0.145013828	0.000057056	0.360691190
K-Means	0.547039923	0.442364664	0.146243498	0.005325219	0.363598977
FCM	0.539683640	0.359194292	0.145013832	0.000000000	0.360587955
GA+K-Means	0.547247236	0.442343813	0.146243505	0.005344237	0.363601128
GA+FCM	0.539596702	0.359203559	0.144987718	0.000057056	0.360663231
PSO+K-Means	0.547247236	0.442343813	0.146243505	0.005344237	0.363601128
PSO+FCM	0.539596702	0.359189658	0.144987718	0.000057056	0.360663231
GTO+K-Means	0.547247236	0.442343813	0.146243505	0.005344237	0.363601128
GTO+FCM	0.539596702	0.359201242	0.144987718	0.000057056	0.360663231
BO+K-Means	0.547247236	0.442343813	0.146243505	0.005344237	0.363601128
BO+FCM	0.539596702	0.359196608	0.144987718	0.000057056	0.360663231
ISGO+K-Means	0.547247236	0.442343813	0.146243505	0.005344237	0.363601128
ISGO+FCM	0.539676953	0.359191975	0.145013828	0.000057056	0.360691190
EO+K-Means	0.547247236	0.442343813	0.146243505	0.005344237	0.363601128
EO+FCM	0.539676953	0.359189658	0.145013828	0.000057056	0.360691190
HO+K-Means	0.547247236	0.442343813	0.146243505	0.005344237	0.363601128
HO+FCM	0.539596702	0.359191975	0.144987718	0.000057056	0.360663231
MetaOOD+K-Means	0.000000000	0.701606019	0.000000000	0.003803728	0.845792655
MetaOOD+FCM	0.539683640	0.720371479	0.145013832	0.000000000	0.599103918
SPM+K-Means	0.571783782	0.758365743	0.122186613	1.000000000	0.829662180
SPM+FCM	0.562421241	1.000000000	0.138000829	0.006466337	1.000000000
GT+K-Means	0.547039923	0.442364664	0.146243498	0.005325219	0.363598977
GT+FCM	0.539683640	0.720371479	0.145013832	0.000000000	0.599103918

Table 6.8 reveals that the proposed SAIFGO significantly enhances clustering performance on the Wave Dataset, which possesses a strong intrinsic structure. A substantial improvement is observed across all clustering validity indices compared to conventional metaheuristic-based hybrid algorithms. Most notably, SAIFGO+K-Means nearly doubles the CH-Score of the standard K-Means approach, while simultaneously achieving the most optimal Silhouette Score and the lowest DB-Index. These results confirm that the SAIFGO optimization is exceptionally effective at capturing well-defined cluster boundaries and maintaining structural density in large-scale datasets with clear patterns.

Chapter 7

Analysis of Proposed Algorithm

7.1 Inspection

The experimental analysis demonstrates that the proposed Improved Fungal Growth Optimization (IFGO) approach provides noticeable improvements in clustering quality, stability and convergence behavior. Across multiple datasets proposed algorithm consistently show stronger performance in terms of cluster validity indices. Furthermore, statistical analyses validate the robustness and statistical significance of the proposed approach. These results confirm that IFGO-based variants often achieve more consistent and statistically distinguishable clustering structures comparatively.

7.2 Advantages

The hybridization of IFGO with K-Means and Fuzzy C-Means (FCM) provides flexibility in clustering. The K-Means variant offers faster computation and clear cluster boundaries, while the FCM variant allows soft clustering. IFGO demonstrates strong robustness across different datasets, including high-dimensional and complex data. To overcome these shortcomings, the present research study aims at introducing a new form of FGO method called Improved Fungal Growth Optimization (IFGO).

The main contributions of this research can be summarized as follows:

- An improved balance between exploration and exploitation phases.

- The integration of chaotic initialization, Levy flight-based mutation and opposition-based learning to enhance global search capability and convergence stability.
- The development of a composite multi-objective fitness function that combines multiple cluster validity indices for robust and balanced clustering.
- A novel hybrid clustering framework that integrates IFGO with K-Means and FCM for effective centroid refinement.
- Extensive experimental and statistical validation on benchmark datasets demonstrating the superiority of the proposed method.

Although clustering has been extensively studied in machine learning and data mining several limitations still exist in traditional and metaheuristic-based clustering approaches. To address these issues there is a need for an improved clustering framework that enhances centroid initialization, strengthens global search capability and provides better robustness across diverse datasets. The proposed Improved Fungal Growth Optimization (IFGO) based clustering approach aims to address these gaps by integrating chaotic initialization, adaptive search strategies, Lévy flight mutation and hybrid clustering mechanisms to achieve more stable clustering performance.

7.3 Limitation

Despite of its improved performance the proposed algorithm has limitations. One of the primary limitations is computational complexity. Since IFGO integrates multiple optimization mechanisms such as chaotic initialization, Lévy flight mutation and opposition-based learning the overall computational cost can become higher compared to traditional clustering methods. Another limitation is the dependence on parameter tuning. The algorithm may face challenges with extremely high dimensional datasets. In such cases, the optimization process may require more iterations to find an optimal solution.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

Improved Fungal Growth Optimization (IFGO) based hybrid algorithms display superior consistency along with multidimensional robustness. While maintaining variance stability and structural integrity in simpler settings IFGO based algorithms provide complete feature level statistical separability. IFGO exhibits reduced feature bias with improved dispersion control and stronger cross domain reliability. The experimental results confirm the efficaciousness of the proposed algorithm for real-world applications. To improve the computational efficiency for large-scale datasets parallel optimization techniques can be introduced with the proposed algorithm. In addition to improving computational efficiency, the proposed IFGO-based clustering approach demonstrates a strong ability to maintain stability across different datasets. IFGO is capable of preventing premature convergence which ensures that the algorithm continues to search for better cluster configurations throughout the optimization process. This indicates another important aspect of the proposed approach. The obtained results indicates improved clustering accuracy and stability compared with others. Integrating the proposed IFGO algorithm with deep learning techniques may provide new opportunities for improving clustering performance in different high dimensional dataset environment. This could enhance the ability of the algorithm to extract meaningful patterns from large and complex datasets. Overall, the findings of this study highlight the effectiveness of the IFGO based clustering approach as a powerful optimization framework for data clustering problems. The proposed method not only improves clustering quality. It provides a flexible and scalable solution for high dimensional dataset. In summary, by introducing improved search capabilities, enhanced

robustness and better adaptability to different types of datasets with FGO contributes to the advancement of optimization based clustering techniques.

8.2 Future Scope

Clustering has a wide range of applications across various dimains. Enhancing clustering techniques can contribute significant and diverse benefits to society. Some of the key application areas and their impacts are presented in the table below.

TABLE 8.1: Usage of Improved Clustering Algorithms Across Domains

Domain	**Usage**	**Impact**
Healthcare	Grouping patients with similar symptoms or medical history	Enables accurate disease diagnosis, personalized treatment planning, and early detection of potential health risks.
Marketing and Business	Customer segmentation based on purchasing behavior	Supports targeted marketing strategies, improves customer satisfaction, and increases business efficiency.
Social Networks	Community detection and grouping users with similar interests	Enhances recommendation systems and improves online user engagement.
Image Processing	Image segmentation and pattern recognition	Applied in medical imaging, facial recognition, and automated object detection systems.
Cybersecurity	Detecting abnormal patterns in network traffic	Helps identify cyber attacks, fraudulent activities, and potential security threats early.
Bioinformatics	Clustering gene expression data	Helps researchers understand genetic diseases and accelerate drug discovery.
Recommendation Systems	Grouping users with similar preferences	Improves recommendation of products, movies, and online content.

Table 8.1 summarizes the wide range of application of clustering.

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